

Kunal Pratap Singh

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SKILLS

- **AI/ML & Data Science:** Deep Learning, Computer Vision, NLP, Transformers, Transfer Learning, Feature Engineering, Model Evaluation, Data Analysis & Visualization
- **Libraries & Frameworks:** PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, FastAPI, Flask, React, Next.js, Node.js
- **Generative AI & LLMs:** Hugging Face, LangChain, RAG Pipelines, Prompt Engineering, LoRA Fine-tuning, Cohere AI, ChromaDB
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, SageMaker), Linux, Cloud Monitoring, Docker & Data Pipelines
- **Programming Languages:** Python, C++, Java, JavaScript, TypeScript, SQL, HTML/CSS
- **Databases & Tools:** MongoDB, Git, Jupyter Notebook, Figma
- **Core Strengths:** Problem Solving, Team Collaboration, Communication, Rapid Learning, Project Development

INTERNSHIP

Indian Space Lab, Remote [June 2025 - July 2025] - Intern

- Analysed drone flight telemetry data using Python (Pandas, NumPy); identified 3 key performance bottlenecks in autonomous navigation algorithms and modelled optimisation strategies for battery life and flight stability.
- Produced a technical report on algorithmic performance evaluation of navigation systems; findings used to guide development priorities.

PROJECTS

DDoS Attack Detection & Mitigation in SDN using DNN: [\[Github\]](#)

Python, PyTorch, Ryu SDN, Mininet, FastAPI

- Achieved 99.65% detection accuracy using a PyTorch DNN with BatchNormalization, Dropout, and class balancing; model deployed via FastAPI for real-time IP-level flow blocking on a live Ryu + Mininet topology.
- Debugged Ryu/eventlet environment incompatibilities on Fedora; resolved OpenFlow feature mismatches for production-grade flow entry matching — demonstrated working live demo with supervisor approval.

AI Interview Coach: [\[Github\]](#) [\[Link\]](#)

Next.js 14, TypeScript, Node.js, MongoDB, LLM API

- Built an LLM-powered interview simulation platform with role-specific system prompts delivering real-time AI feedback; average API response latency under 2s per session.
- Implemented progress tracking, multi-role practice scenarios, and JWT + Google OAuth auth; full-stack: Next.js 14, Express, MongoDB, Tailwind, Framer Motion.

Smart Attendance System : [\[Github\]](#)

Python, OpenCV, Flask

- Built a real-time facial recognition attendance system using OpenCV + Flask, reducing manual tracking effort by ~80% in a 70-student test environment.
- Designed an admin dashboard with CSV export and daily/weekly reporting capabilities.

Email Generator : [\[Github\]](#)

FastAPI, Cohere AI

- Engineered an AI-powered ABM email system using FastAPI + Cohere AI with A/B testing and multilingual personalisation; capable of generating 100+ campaign variants at scale.

EDUCATION

VIT Bhopal University, Bhopal, India - B.Tech in Computer Science

[2023 - 2027]

- **8.36 GPA/CGPA***

Jyoti Senior Secondary School, Rewa, India

- **Class XII [July 2022] - 80%**
- **Class X [July 2020] - 90.3%**